

LOGIC GATES

- There are in common 6 gates in 0-levels computer science.

- NOT 

- OR 

- AND 

- NAND 

- NOR 

- XOR 

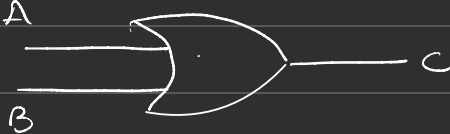
WORKING

we deal in Binary only

① NOT $\text{in}_0 \rightarrow \text{out}_0$

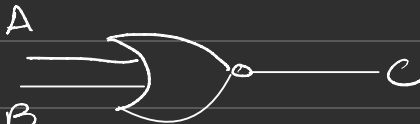
A	B
0	1
1	0

② OR gate



A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

③ NOR



A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

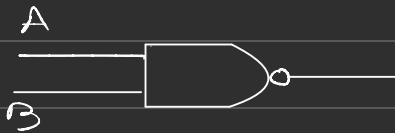
AND

A	B	C
0	0	0
0	1	0
1	0	0
1	1	1



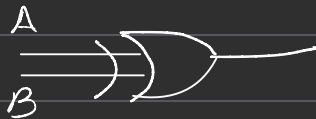
NAND

A	B	C
0	0	1
0	1	1
1	0	1
1	1	0



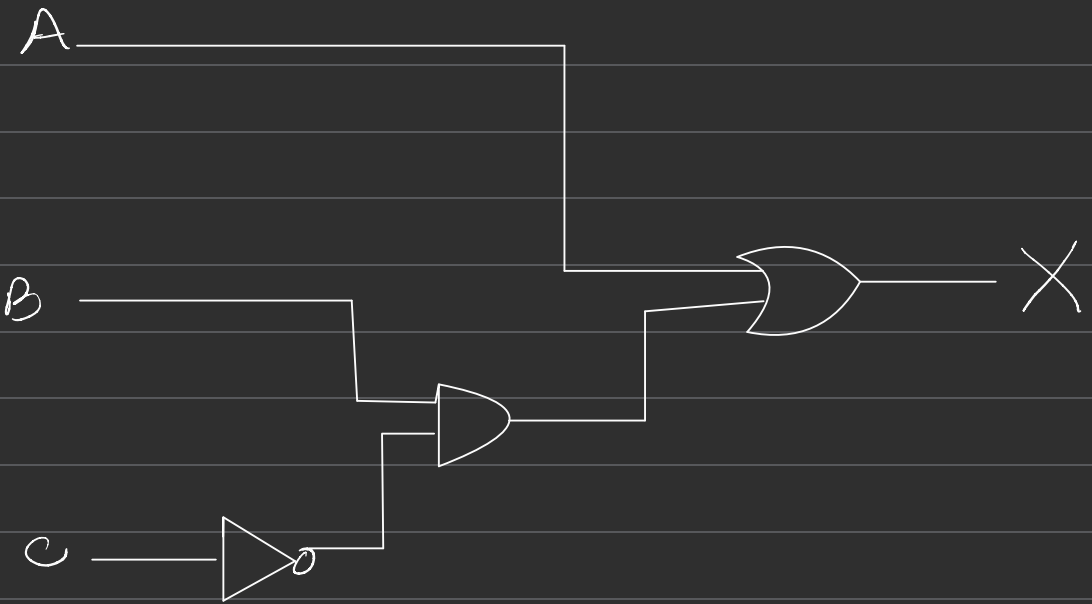
XOR

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0



Question Draw the logic circuit for

$$X = A \text{ OR } (B \text{ AND } \text{NOT } C)$$



$$X = A \text{ OR } (B \text{ AND NOT } C) \text{ OR } (A \text{ AND } B)$$

